BHARAT AI-SoC STUDENT CHALLENGE PROJECT

Diya – Offline Hindi Voice Assistant using Raspberry Pi

TEAM MEMBERS

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HINDI VOICE ASSISTANT DIYA

1. Introduction

Voice assistants have become an important part of modern human–computer interaction. Most existing assistants such as Alexa, Google Assistant, and Siri depend heavily on cloud connectivity, which raises concerns related to privacy, latency, and internet dependency.

This project presents **Diya**, an *offline Hindi voice assistant* developed using a **Raspberry Pi** platform. The assistant operates completely without internet access for core features and performs **speech recognition**, **intent processing**, and **speech synthesis** locally.

The system is designed to:

- Understand Hindi voice commands
- Operate without cloud services
- Provide real-time responses
- Control local functionalities such as music playback and alarms

Objective summary: Build a privacy-preserving, low-cost, real-time offline assistant suitable for embedded systems.

2. Objectives

The primary objectives of this project are to:

- Develop an offline Hindi speech recognition system
- Implement wake-word based activation
- Design an intent recognition engine
- Generate natural Hindi speech responses
- Optimize execution for Raspberry Pi hardware
- Enable multimedia control and scheduling features

3. System architecture

Diya follows a modular pipeline architecture:

```
flowchart TD
    A["User Speech"] --> B["Microphone Input"]
    B --> C["Audio Processing & Resampling"]
    C --> D["Vosk Speech Recognition"]
    D --> E["Wake Word Detection"]
    E --> F["Intent Parsing Engine"]
    F --> G["Action Execution / AI Response"]
    G --> H["Piper Text-to-Speech"]
    H --> I["Audio Output (Speaker/Earbuds)"]
```

4. Methodology

4.1 Audio acquisition

Audio input is captured using Bluetooth earbuds or a microphone through the `sounddevice` library at a **48 kHz** sampling rate.

The signal is resampled to **16 kHz** using:

```
scipy.signal.resample_poly
```

This matches the Vosk ASR model requirements.

4.2 Speech recognition (ASR)

Offline speech recognition is implemented using:

- **Vosk Hindi Model (vosk-model-hi-0.22)**

Key features:

- Fully offline operation
- Low CPU usage
- Real-time decoding

- Robust Hindi recognition

4.3 Wake word detection

The assistant activates only when the wake word **"दीया"** is detected.

To improve robustness:

- Multiple phonetic variations are supported
- Fuzzy matching using `SequenceMatcher` handles pronunciation errors

4.4 Command listening mechanism

After wake word detection:

1. Music playback pauses.
2. Audio buffer is cleared.
3. The assistant listens for commands for a fixed timeout period.
4. Silence detection determines speech completion.

This prevents accidental command triggering.

4.5 Intent recognition

A rule-based intent engine analyzes the recognized text.

Supported intents include:

- Greeting
- Time and date queries
- Alarm setting
- Music control
- Next/previous song
- Stop playback
- Exit command

Keyword grouping improves recognition reliability despite ASR spelling variations.

4.6 Text-to-speech (TTS)

Speech output is generated using:

- **Piper Neural TTS**
- Hindi voice: *Priyamvada*

Pipeline:

Text → Piper → Raw Audio → `paplay` → Speaker

Advantages:

- Natural speech quality
- Offline neural synthesis
- Low latency

4.7 Music playback module

Music is managed using the **VLC command-line interface**:

- Playlist auto-loading
- Pause during listening
- Resume after response
- Thread-safe process handling

5. Advantages

- Fully offline operation
- Privacy preserving
- Low-cost implementation
- Hindi language support
- Real-time performance
- Extendable architecture

6. Limitations

- Limited conversational intelligence without AI fallback
- Dependent on ASR accuracy
- Noisy environments may reduce recognition accuracy

7. Future enhancements

- Integration with a local LLM (Ollama)
- Smart home IoT control
- Continuous conversation mode
- Speaker identification
- Noise suppression using DSP techniques

8. Conclusion

The project successfully demonstrates an efficient offline Hindi voice assistant running on embedded hardware. By combining offline ASR, rule-based intent recognition, and neural TTS, the system achieves real-time interaction without cloud dependency.

Diya shows that intelligent voice interfaces can be implemented locally while maintaining privacy, affordability, and responsiveness.

9. References

- Vosk Speech Recognition Toolkit
- Piper Text-to-Speech Engine
- Raspberry Pi Documentation
- VLC Media Player CLI Documentation
- Python SoundDevice Library